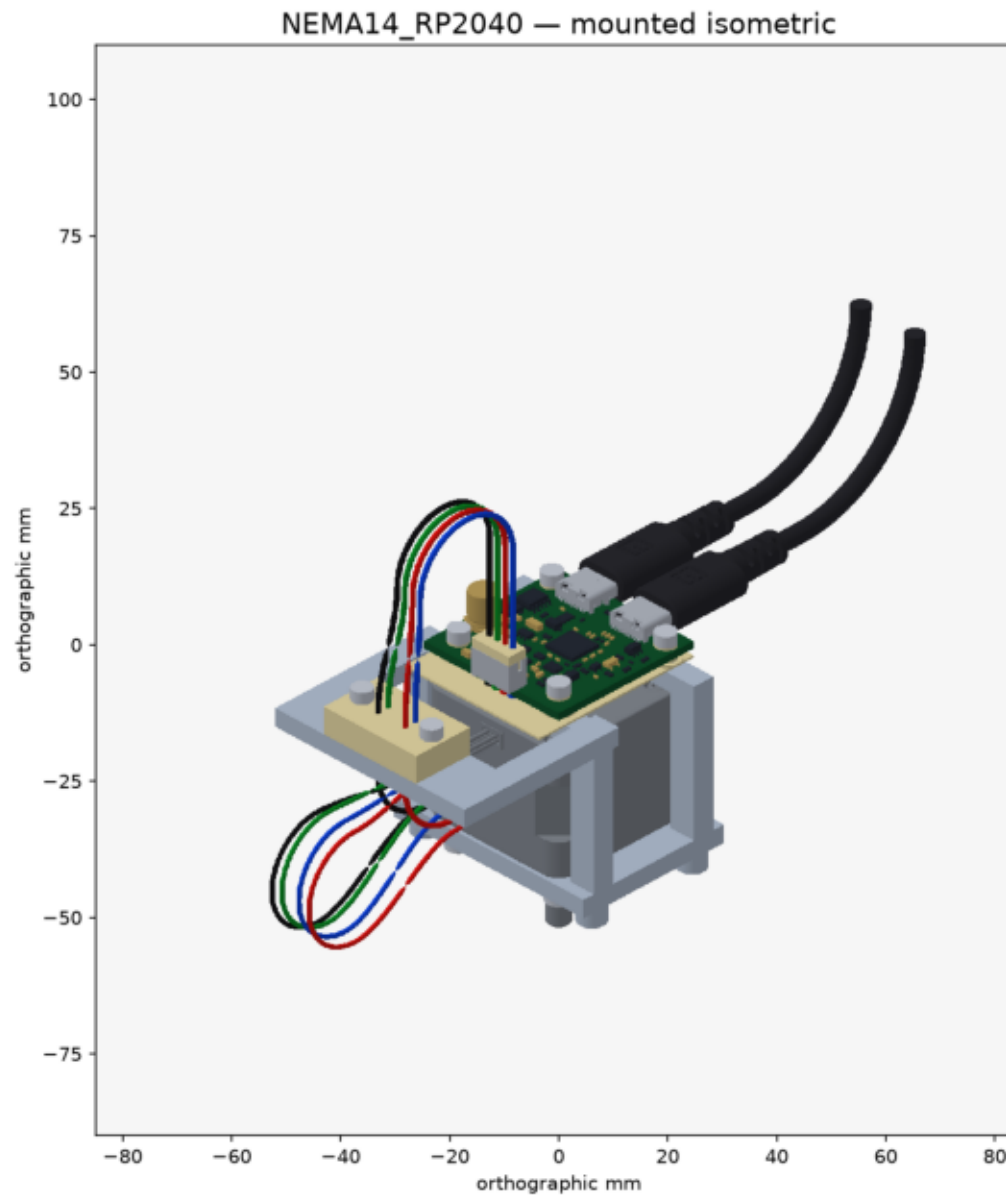
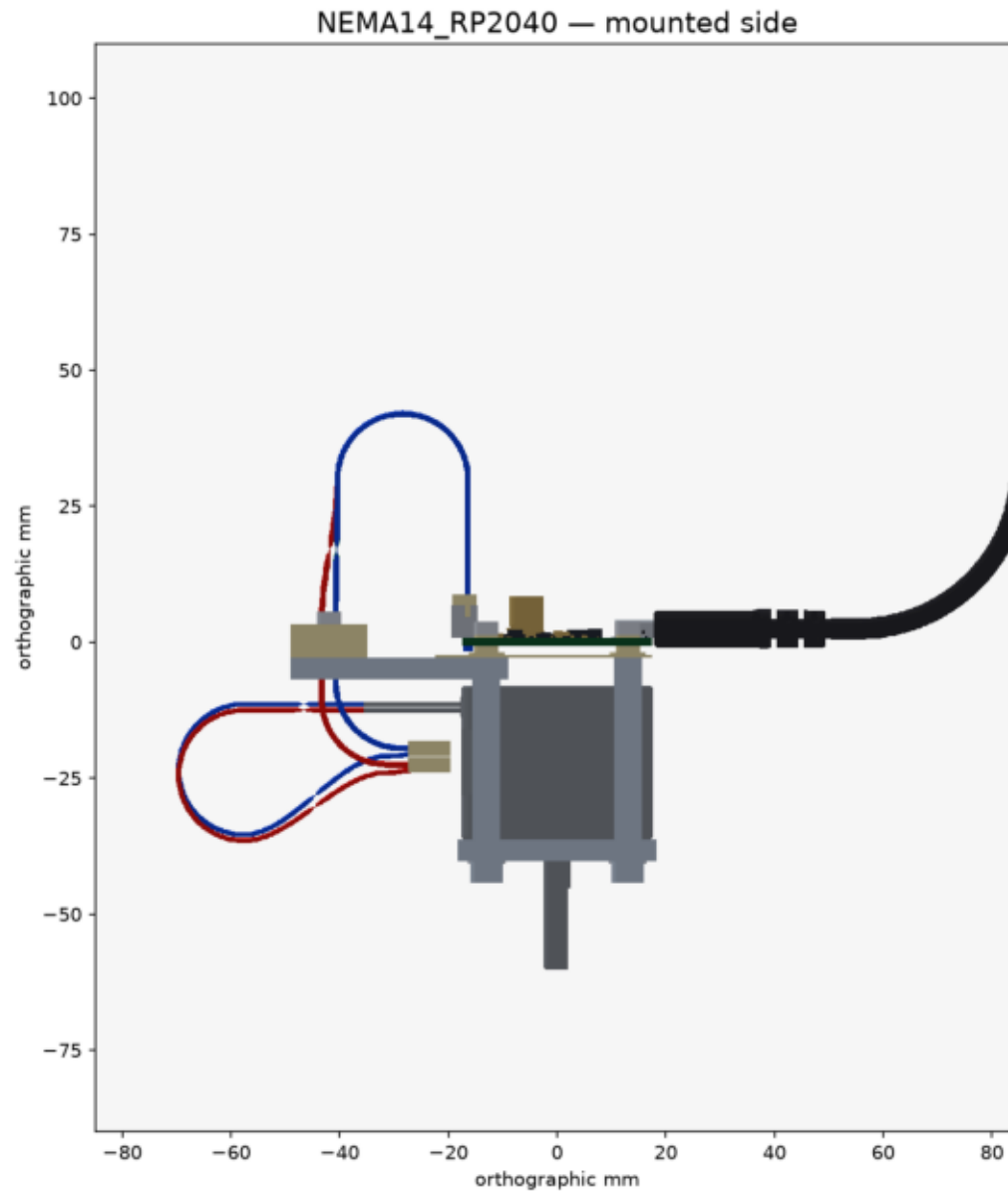


NEMA14_RP2040 — HAR-001 / HAR-002 installation



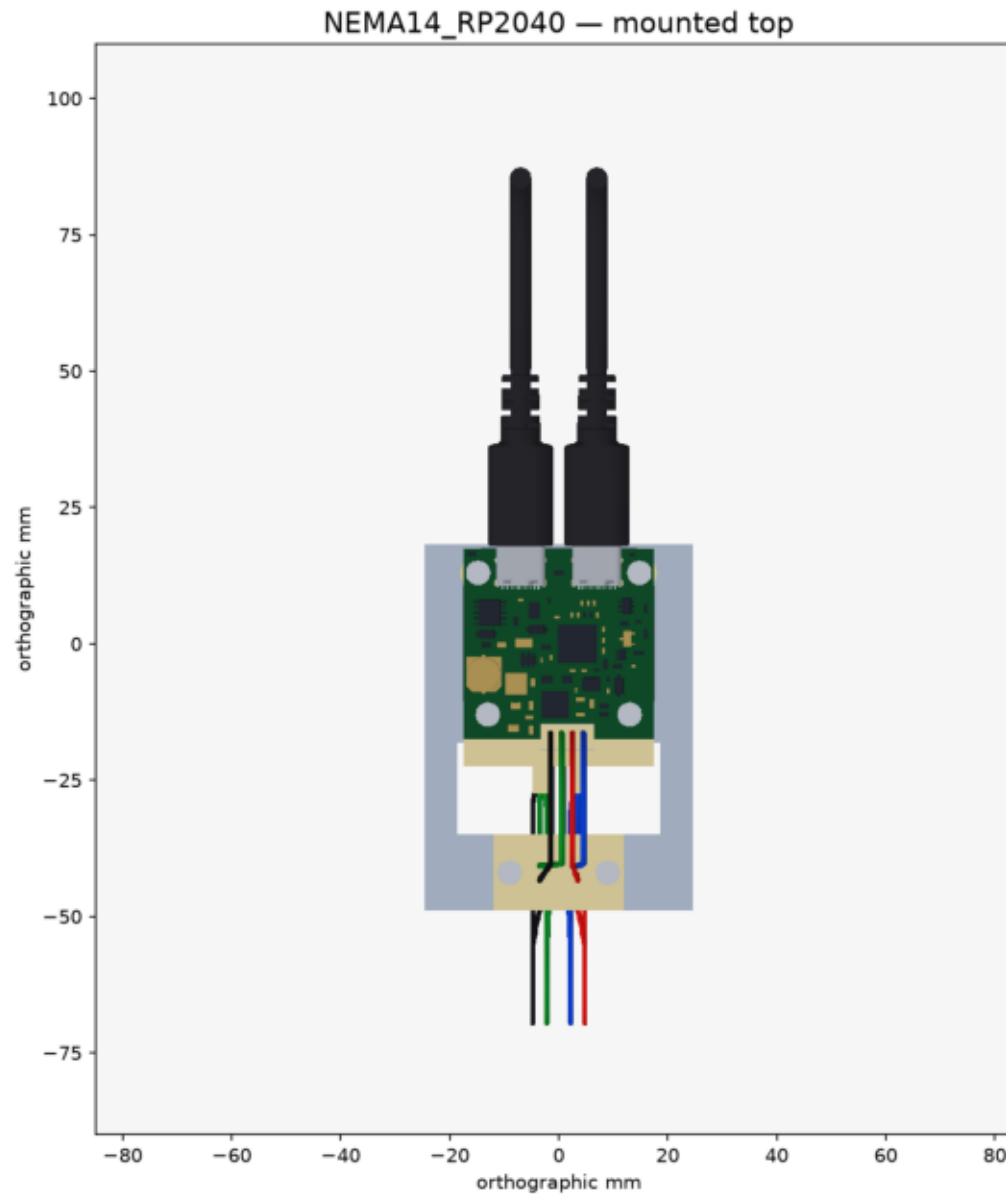
Nominal installation envelopes; factory key/contact and lead stock qualification pending. No measured fit.

NEMA14_RP2040 — HAR-001 / HAR-002 installation



Nominal installation envelopes; factory key/contact and lead stock qualification pending. No measured fit.

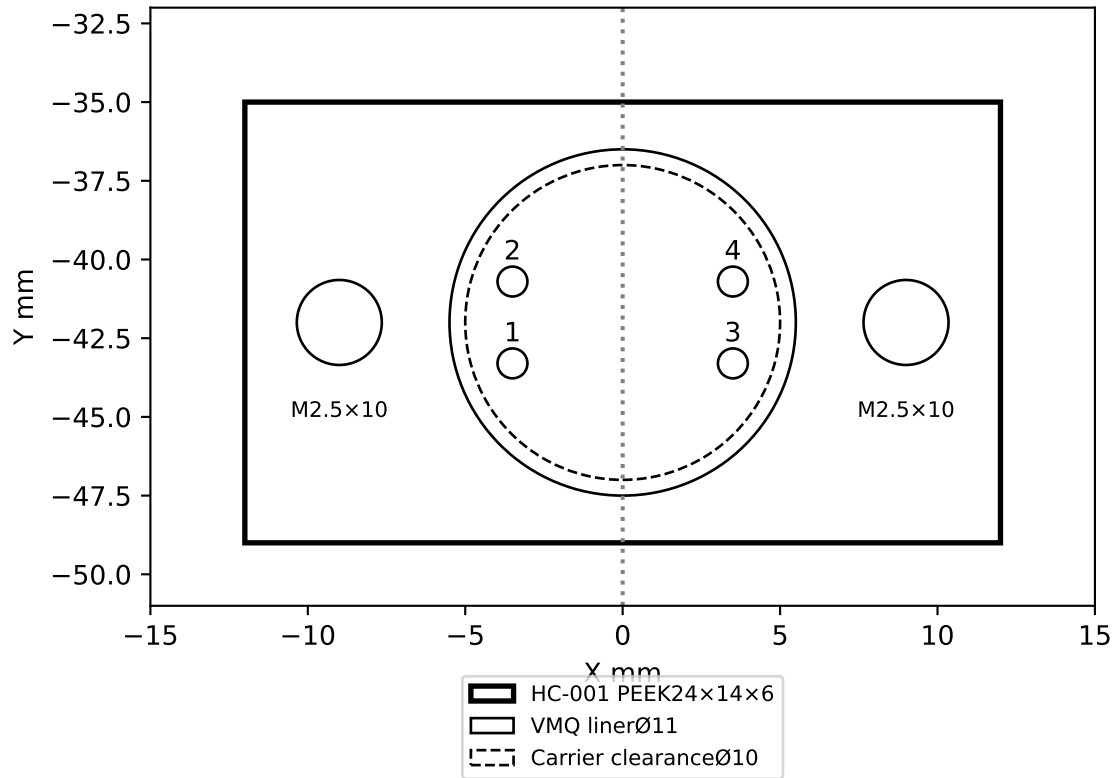
NEMA14_RP2040 — HAR-001 / HAR-002 installation



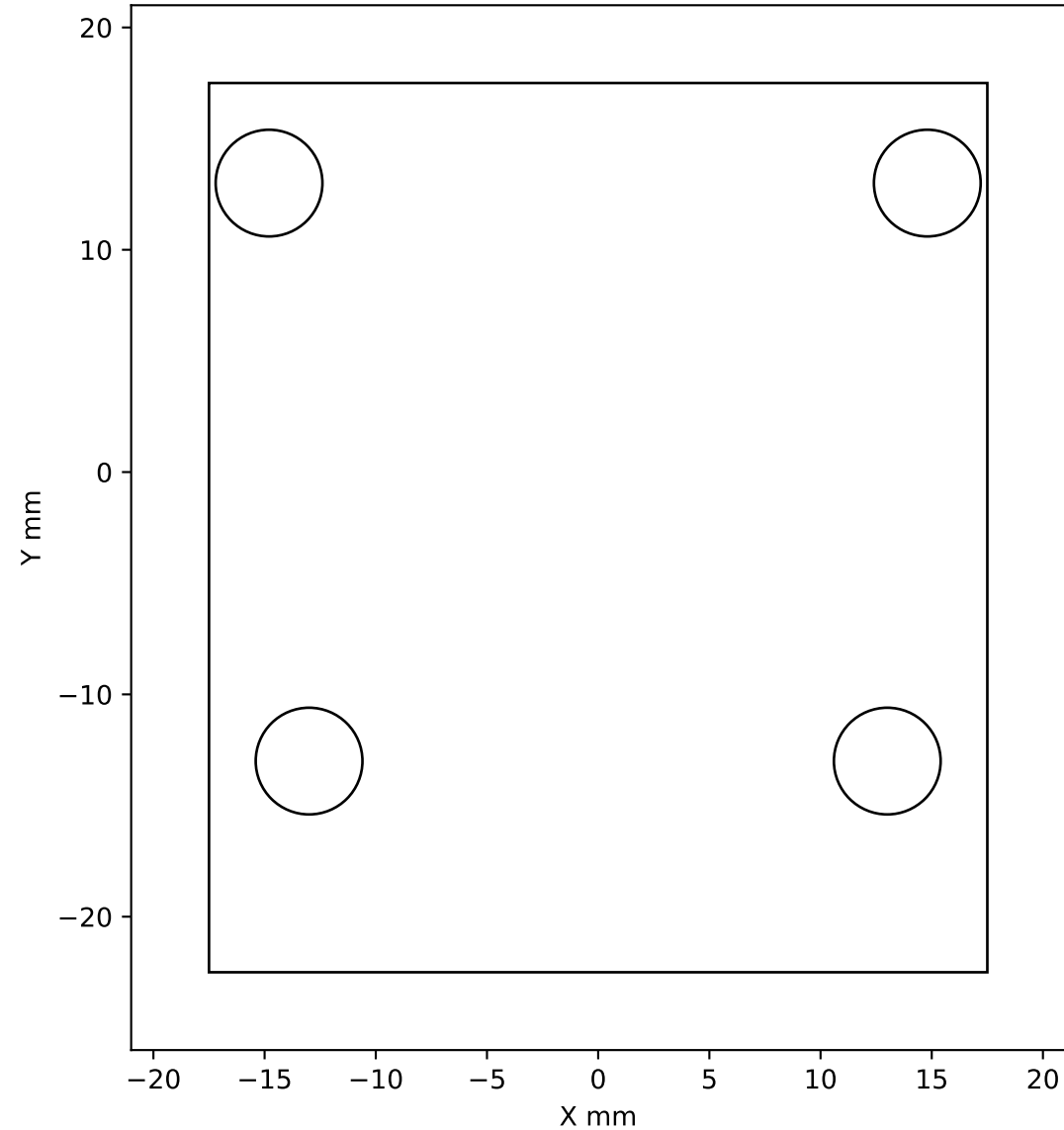
Nominal installation envelopes; factory key/contact and lead stock qualification pending. No measured fit.

HC-001 / INS-001 manufacture and assembly datums

Clamp top datum, mm; X split halves



Barrier pattern, 4xØ4.8; sheet center Y-2.5



Clamp bores $\varnothing 0.95 \pm 0.05$; accepted wire OD 1.05..1.20. Seat Z-2.8; carrier ± 0.05 . Certificates/preload/pull-test pending.

Assembly specification and host requirements

Front carrier: 49.2×36.4×4mm; 6061-T6/T651 ±0.05mm.

Front motor 4×M3 at 26±0.2 square, depth ≥4mm; never use rear screws.

Carrier 4×Ø4.1±0.05 at 26±0.05; Ø22.15±0.05 locating bore.

Pilot root relief Ø23.4±0.05×0.40±0.05 deep from motor mating face.

Motor fasteners 4×M3×8 +0.5mm washers; bounded engagement 3.2..3.8mm.

Host 4×M3 at X±21.6, Y±13; 6mm available depth, engage 4.5..5.5mm.

Host front face Z−44.2; shaft projection beyond host 14.9..17.1mm.

PCB 4×M2.5×8; PEEK top 0.5/lower 1mm; nominal metal engagement 4.9mm.

Motor PCB header: JST B4B-PH-K-S(LF)(SN), C131334; bore 0.75±0.05.

Mate PHR-4 +4×SPH-002T-P0.5S; AWG 24..30, OD 0.8..1.5mm.

1 Black A+/OUT1A; 2 Green A−/OUT1B; 3 Red B+/OUT2A; 4 Blue B−/OUT2B.

Belden 83004: 010100black /005100green /002100red /006100blue.

Extension nom OD 1.1, PTFE 200°C, min static bend 11mm; modeled 12mm.

HAR-002 factory lead OD/AWG/material unknown: inspect before cutting.

Individually insulated/staggered laps splices in bounded 2.8×8×2.8 envelopes.

Splice insulation candidate RT-375-1/8-X; certificate/process qualification pending.

Tail projection ≤0.8, fillet ≤0.5 below PCB; qualified temperature-rated tailcaps.

INS-001 Nomex 4100.51; ≥150°C material certificate required.

HC-001 PEEK/VMQ 50±5 Shore A, 2×M2.5×10; 4mm nominal metal engagement.

Header withdrawal ≥10mm, gripping housing; deenergize/discharge before unplug.

PD Tensility 10-06137; DATA Tensility 10-06139; maximum body 12.5×32.5×7.6.

USB pitch 14mm; worst placement body separation 1.3mm; reserve R30 bend.

R30 is host space, not manufacturer-qualified cable bend/fatigue radius.

Complete reserved bbox (mm): [-24.602, -70.082, -60.211, 24.602, 87.399, 42.559]

Minimum host envelope (mm): [49.204, 157.48, 102.771]